Project Management Plan for Theia

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Team 10: {Placeholder}

9/20/26

Project Management Plan for Theia

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Revision History

Name	Timestamp	Reason for Changes	Version
Alex Shaw, David Del Val Fraire, Seth Trimberger, Danbi Kim, Johanna Siebol	9/10/26 – 8:00 PM	Initial edits to the template.	0.0.0
Alex Shaw	9/10/26 – 8:45 PM	Started sections 4.1 and 4.2.	0.0.1
Johanna Siebol	9/10/26 – 8:45 PM	Started section 1.1.	0.0.2
Seth Trimberger	9/15/26 – 8:45 PM	Started section 3.1.	0.0.3
David Del Val Fraire	9/15/26 – 10:00 PM	Started sections 2.1, 2.2, and 2.3.	0.0.4
Danbi Kim	9/16/26 – 9:30 AM	Started sections 1.3, 1.4, and 1.5.	0.0.5
Seth Trimberger	9/16/26 – 4:45 PM	Finished section 3.1 and started section 3.2.	0.0.6
Johanna Siebol	9/20/26 – 12:00 PM	Updated sections 1.1 and 1.2.	0.0.7
Seth Trimberger	9/20/26 – 1:45 PM	Finished section 3.2 and 3.3.	0.0.8
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Danbi Kim	9/20/26 – 9:50 PM	Updated sections 1.4 and 1.5.	0.1.1
David Del Val Fraire	9/20/26 – 10:06 PM	Updated sections 2.1, 2.2, and 2.3.	0.1.2
Seth Trimberger	9/20/26 – 10:11 PM	Updated section 3.1, 3.2 and 3.3	0.1.3
Johanna Siebol	9/20/26 – 10:45 PM	Updated section 1.1 and 1.2.	0.1.4

1. Overview

1.1. Project Purpose, Objectives, and Success Criteria

This project aims to create an application named *Theia* to assist blind and low-vision individuals in locating rooms and other significant locations within a building. The application will use smartphone speakers, microphones, and accessibility features to provide navigation instructions and safety information, including the ability to contact emergency services when needed.

The system assumes that rooms and other significant locations are equipped with wireless beacons capable of communicating their location to the smartphone. Due to project resource constraints, the functional prototype will simulate receiving these beacon signals rather than require physical beacon deployment. The application will use the simulated location information to guide the user between indoor locations.

The project will be considered successful if the prototype demonstrates simulated indoor navigation between locations, provides accessible user interaction consistent with applicable WCAG 2.2 guidelines, and supports the required emergency dial-out functionality.

The scope of the project is limited to indoor navigation between rooms or other significant locations. Navigation within individual rooms is outside the project scope. The system will account for static obstacles but will not account for moving obstacles such as pedestrians.

1.2. Project Deliverables

Deliverable	Delivery Date	Delivery Method	Comments
Project Phase I: Preliminary Project Plan	9/20/26	Word Document	
User Manual	10/4/26	Word Document	
Interface Mock-Up / Prototype	10/4/26	PowerPoint/Word Document	
WRS Document	10/6/26	Word Document	
AS-IS/TO-BE scenarios	10/8/26	PowerPoint/ YouTube Video	
Phase I Demo Slides	10/8/26	PowerPoint	
Project Phase I: Peer Review	10/12/26	Word Document	This is required of all members.
GenAI Usage and Verification Report	10/18/26	Word Document	
Project Phase I: Final Submission	10/18/26	Word Document	
Project Phase II: Final Submission	TBD	Canvas/Github/Word Document	This is due around the 13th or 14th week of the semester.

1.3. Assumptions, Dependencies, and Constraints

The timeline is fixed by the course deadlines for the deliverables. The project operates with no budget and relies exclusively on free software, mock-up, and tools provided by the school. Also, the team personnel are fixed by student course enrollment and assigned team. No additional personnel recruitment is allowed. Thus, schedule, cost, and staff are the constraints of this project. High analytical rigor in requirements elicitation, seamless domain-to-specification traceability in the WRS document, and accessibility for blind and low-vision users drive project success. Therefore, quality is a primary success driver for this project. Finally, the functional scope of the TO-BE system and prototype can be adjusted based on the team's design choices and the minimum mock-up requirements. Thus, the features are the degrees of freedom.

Assumptions

- AS-1: Target users have access to a smartphone with standard technologies and hardware sensors (such as screen readers, voice control, and motion sensors).
- AS-2: Users have baseline familiarity with utilizing smartphone audio/haptic feedback or screen-reading tools for mobile interactions.
- AS-3: GenAI tools used by the team will remain accessible, functional, and permitted throughout the semester schedule.

Dependencies

- DE-1: Team collaboration infrastructure including GitHub repository, video hosting services such as YouTube, and group communication platforms.
- DE-2: Course materials and evaluation rubrics.
- DE-3: Accessibility guidelines and interface standards (e.g., WCAG) required to evaluate mock-ups and scenarios.

Constraints

- CO-1: AS-IS/TO-BE presentation, WRS document, prototype and manual, project plan, AI and teamwork report must be submitted by the deadlines of the course schedule.
- CO-2: The AS-IS/TO-BE presentation must adhere to a 10-15 minute recording and should be uploaded as an unlisted YouTube video, following the mandatory scenario format.
- CO-3: All GenAI usage must be documented with verification evidence and presented in a report.

1.4. References

World Wide Web Consortium (W3C). *Web Content Accessibility Guidelines (WCAG) 2.2.* Available at: https://www.w3.org/TR/WCAG22/.

School of Electrical Engineering and Computer Science, Washington State University. (2026). *Cpt S 484 Team Project Specification: Phase I - Requirements Elicitation (v2.0).*

Apple Inc. *Human Interface Guidelines: Accessibility.* Apple Developer Documentation. Available at: https://developer.apple.com/design/human-interface-guidelines/accessibility.

Apple Inc. *Accessibility on IOS (UIAccessibility).* Apple Developer Documentation. Available at: https://developer.apple.com/documentation/accessibility.

1.5. Definitions and Acronyms

- AI: Artificial Intelligence. The simulation of human intelligence processes by computer systems.
- AS-IS Scenario: A narrative description of how users currently complete tasks or navigate challenges without the proposed software solution.
- TO-BE Scenario: A narrative description illustrating how a user will complete indoor navigation tasks using the proposed Theia application.
- GUI: Graphic User Interface. A visual way of interacting with a computer or smartphone through graphical icons and visual indicators.
- NFR: Non-Functional Requirement. A specification that describes system attributes, quality characteristics, and operational constraints rather than specific behaviors.
- WRS: A requirements engineering framework that documents domain context, stakeholder goals and objectives, and detailed software capabilities.
- WCAG: Web Content Accessibility Guidelines. Technical benchmarks developed by the W3C outlining internationally recognized success criteria for accessible digital content and interactions.
- UI/UX: User Interface/User Experience. The visual, auditory, and haptic touchpoints of software combined with the overarching usability and emotional satisfaction of the user interacting with it.
- TTS: Text-to-Speech. The algorithmic synthesis of human speech output from digital text to convey wayfinding cues and situational warnings.
- iOS: The mobile operating system developed by Apple Inc. for iPhone hardware, providing foundational accessibility layers like VoiceOver.
- AR: Augmented Reality. An interactive experience where digital information is merged in real time with the user's physical environment.
- CV: Computer vision. A domain of artificial intelligence enabling computational agents to extract, process, and interpret meaningful information from digital imagery or sensor feeds.

2. Project Organization

2.1. Process Model

We will be using an iterative and incremental process model to develop this project, with phases organized around the deliverable dates listed in Section 1.2. Each phase produces one or more artifacts that are reviewed internally before submission. Documentation will be produced in accordance with the formatting required for coursework.

The key phases and their associated tasks are:

Phase 1: Project Initiation and Planning (through 9/20)

- Establish team structure, communication channels and shared repositories.
- Define project purpose, objectives, and measurable success criteria
- Identify assumptions, dependencies, and constraints
- Enumerate deliverables and assign ownership
- Produce and baseline the Preliminary Project Plan

Phase 2: Requirements Elicitation, Domain Analysis, and Interface Design (9/20 - 10/6)

- Conduct domain research on indoor navigation for blind and low-vision users
- Review accessibility standards (WCAG 2.2) and mobile accessibility interface conventions
- Develop the interface mock-up to the minimum required fidelity (due 10/4)
- Produce the user manual against the mock-up (due 10/4)
- Elicit stakeholder goals and derive functional and non-functional requirements
- Establish traceability from domain context through to software capabilities
- Produce the WRS document (due 10/6)

Phase 3: Scenario Modeling and Peer Review (10/6 - 10/12)

- Author AS-IS scenarios describing how target users complete indoor navigation tasks today
- Author TO-BE scenarios describing task completion using Theia
- Record and publish the AS-IS/TO-BE presentation within the required format and duration (due 10/8)
- Complete individual peer reviews (due 10/12)

Phase 4: Final Review and Integration (10/12 - 10/18)

- Incorporate peer review and instructor feedback into all Phase I artifacts
- Run the cross-document consistency and pre-submission conformance check (dates, owners, cross-references)
- Finalize the GenAI Usage and Verification Report
- Update the table of contents, page numbers, and revision history
- Baseline and submit the Phase I final submission (due 10/18)

Within each phase the team follows the same working cycle: plan the deliverable into assignable work items, produce the assigned sections, review each artifact with at least one team member other than the author, and integrate the reviewed work into a new baselined version. No artifact is submitted without being reviewed.

2.2. Organizational Structure

The members involved in developing this project are Alex Shaw, David Del Val Fraire, Seth Trimberger, Danbi Kim, and Johanna Siebol. The team does not have a management hierarchy. Decisions are made by the team collectively.

Every member holds one primary role and contributes to all major deliverables. Where a deliverable spans multiple areas, the deliverable owner listed in Section 2.3 has final integration authority for that artifact.

All project communication takes place in shared channels visible to every member. Communication with the instructor and course staff is handled by the Communication Lead, with responses relayed to the full team.

Interface	**Participants**	**Channel**	**Frequency**
Internal coordination	All members	Discord (text channels)	Ongoing
Working sessions	All members	Discord (voice), Zoom, and in-person meetings	Weekly
Status Updates	All Members	Discord (text threads)	Weekly
Document collaboration	All members	Google Drive	Ongoing
Version control and artifacts	All members	GitHub repository	Ongoing

Instructor and course staff	Communication Lead on behalf of team	Course-designated channel and email	As needed
Peer review exchange	Deliverable owners and reviewing team	Course-designated channel (Canvas)	Per peer review milestone

Engineering support functions are held as assigned roles:

- **Configuration Management**: The Repository and Configuration Lead owns the repository structure, file naming and version numbering. Artifacts are versioned in the team GitHub repository, with each baselined deliverable committed to the main branch and tagged with its version number.
- **Quality Assurance**: The Repository and Configuration Lead owns template conformance, cross-document consistency, and the pre-submission checklist.
- **Verification and Validation**: The Accessibility and Design Lead owns validation of the mock-up and scenarios against accessibility standards and the needs stated in the requirements.
- **Process Improvement**: The full team holds a retrospective review after each deliverable submission, with any agreed change recorded in the revision history of this document.

2.3. Roles and Responsibilities

All team members serve as Requirements Analysts and Contributing Authors on all deliverables. The following roles are assigned in addition, for the duration of the project.

Role	Member	Responsibilities
Project Manager / Communication Lead	Alex Shaw	Maintains the project schedule and deliverable tracker; assigns and tracks work items; runs team meetings and records decisions; serves as the single point of contact with the instructor and course staff; owns this Project Management Plan and its revision history; escalates schedule risk.
Accessibility and Design Lead	David Del Val Fraire	Owns the accessibility posture of the product; validates all interface work against WCAG 2.2 and mobile accessibility conventions (DE-3); leads mock-up and prototype development; owns non-functional requirements relating to accessibility and usability; performs validation of the TO-BE design against stated user needs.

Requirements Lead / Systems Analyst	Danbi Kim	Owns the WRS document structure and content; leads requirements elicitation and domain analysis; maintains traceability from domain context to stakeholder goals to software capabilities; adjudicates requirement conflicts and records their resolution.
Scenario and User Experience Lead	Johanna Siebol	Owns the AS-IS and TO-BE scenarios and their conformance to the mandatory scenario format (CO-2); owns the user manual; leads production and recording of the AS-IS/TO-BE presentation; ensures scenario narratives remain consistent with the requirements and the mock-up.
Repository and Configuration Lead / Quality Assurance	Seth Trimberger	Owns the GitHub repository structure and the shared document workspace; enforces version numbering and naming conventions; maintains the GenAI usage log and verification evidence required by CO-3; runs the pre-submission conformance check on every deliverable; and maintains backups of all project artifacts.

Each deliverable has one owner and at least one reviewer. The owner is responsible for integration and on-time submission. The reviewer reads the complete artifact before submission and records findings.

Deliverable	Date	Owner	Reviewer(s)
Project Phase I: Preliminary Project Plan	9/20/26	Alex Shaw	All Other Members
User Manual	10/4/26	Johanna Siebol	David Del Val Fraire
WRS Document	10/6/26	Danbi Kim	David Del Val Fraire, Seth Trimberger
AS-IS / TO-BE Scenarios	10/8/26	Johanna Siebol	Danbi Kim, Alex Shaw
Interface Mock-up / Prototype	10/4/26	David Del Val Fraire	Johanna Siebol, Seth Trimberger
Phase I Demo Slides	10/8/26	Johanna Siebol	Alex Shaw, Danbi Kim
Project Phase I: Peer Review	10/12/26	Each Member	All Members
GenAI Usage and Verification Report	10/18/26	Seth Trimberger	Alex Shaw

Project Phase I: Final Submission	10/18/26	Seth Trimberger	All Other Members
Project Phase II: Final Submission	TBD	TBD at Phase II planning	All Other Members

Roles may be reassigned by team consensus if workload proves unevenly distributed or a member becomes unavailable, with any reassignment recorded in the revision history of this document. For each role, at least one other member maintains enough familiarity with the work to take over as needed.

3. Managerial Process Plans

3.1. Management Objectives and Priorities

The primary management objective of this team is to complete all required deliverables and work for the THEIA project on schedule while making sure the workload is distributed fairly among all team members. Each member will have clearly assigned responsibilities so that everyone contributes a meaningful share of the project. As outlined in Section 2.3 each deliverable has an assigned owner and reviewer and roles can and will be adjusted if workloads become uneven or a team member becomes unavailable.

Another major priority is that we maintain clear and consistent communication throughout the project. Our team will regularly review completed work, discuss current tasks and bring up any issues that will affect upcoming deadlines. The communication methods and meeting expectations used by the team are defined in Section 2.2 which includes using Discord for all ongoing coordination and project updates and weekly meetings will be held through Discord or in person with additional meetings scheduled as needed.

The team will also prioritize accountability and staying on schedule by ensuring each team member is responsible for completing their assigned work by the agreed upon deadlines and ensure they communicate early if they expect a delay. As outlined in Section 2.1 project work is divided up into assignable tasks and will be reviewed by at least one other team member before being integrated and submitted.

Lastly, major project decisions will be made collectively and important deliverables will be reviewed by the team before submission. This follows the organizational structure that is described in Section 2.2 where decisions are made collectively and all members are expected to contribute to all the major deliverables.

3.2. Assumptions, Dependencies, and Constraints

The team assumes that all members will remain actively involved throughout the entirety of the project, contribute a fair share of the work, and communicate early if an issue could affect their work and responsibilities. We also assume that team members may need time to learn and better understand parts of the THEIA project as work progresses, including requirements, engineering concepts, accessibility expectations, and the specific areas assigned to each member.

This project depends on consistent communication, participation and access to the tools and course resources needed to complete the work. As described in Section 2.2 Discord will be used for ongoing team coordination, meetings will be held at least weekly through Discord or in person, Google Drive will be used for document collaboration, and GitHub will be used for version control and project artifacts. The team also depends on course materials, assignment instructions, and instructor feedback to make sure the project continues to meet course expectations.

From a management perspective one of the main constraints is team availability. Since this is an online course, team members have different classes, work, and schedules which can make it difficult to find meeting times that work for everyone. The team will need to plan meetings in advance and rely on communication through Discord which is outlined in Section 2.2 when members are not able to meet in person.

Another constraint is the fixed length of the academic term and the deadlines established by the course. The team must complete the required deliverables within the available semester timeline. Further as described in Section 2.3 if a team member becomes unavailable or the workload becomes uneven, responsibilities can and will be reassigned by the team so that progress can continue and deadlines can still be met.

3.3. Risk Management

The team will identify and review project risks throughout the project. During team meetings risks will be discussed and evaluated based on how likely they are to occur and the impact they could have on the project. If a risk becomes more likely or begins to have a greater impact the team will review the situation and determine what changes need to be made. Identifying these risks early will give the team more time to respond before they cause delays or affect major project deliverables.

Risk	Likelihood	Impact	Response / Control
Missed Deadlines	Medium	High	Use internal deadlines and review progress during our weekly meetings on Discord.
Uneven workload	Medium	Medium	Review task distribution and reassign work when necessary to keep workloads balanced.

Team member unavailable	Medium	High	Communicate the issue early and redistribute important work among available team members.
Communication problems	Medium	High	Maintain regular Discord communication and attend all meetings to ensure the project continues making progress and issues are visible to the team.
Misunderstanding requirements	Medium	High	Review project instructions and course materials as a team and ask the instructor for clarification when needed.
Scheduling conflicts	High	Medium	Plan meetings in advance to ensure everyone is available and change meetings as needed
Loss or conflicting versions of work	Low	High	Store project files in shared locations and use GitHub version control and consistent naming practices to prevent files from being lost or overwritten.
Difficulty learning unfamiliar concepts	Medium	Medium	Allow time for research and share findings, resources, and clarifications with the rest of the team.

4. Technical Process Plans

4.1. Methods, Tools, and Techniques

Theia is planned as a native iOS application for iPhone supporting indoor navigation for blind and low-vision users. Navigation is limited to movement between rooms or significant building locations rather than navigation within individual rooms. Static obstacles may be considered; moving obstacles such as pedestrians are outside the project scope.

The anticipated implementation stack is Xcode and Swift, using Apple accessibility and device frameworks where appropriate. An iOS-first approach reduces device variability and provides direct support for VoiceOver, audio guidance, haptic feedback, and other accessibility features. Core operation is intended to remain on-device without requiring a cloud backend.

The system assumes that rooms and other significant locations contain wireless beacons capable of communicating their location to the smartphone. As the user moves through a building, the application may receive signals such as *Room 301* followed by *Room 302*. The functional prototype will simulate receiving these beacon signals rather than require deployment of physical beacons. Emergency dial-out will also be treated as a required capability.

Phase I tools will include Figma or equivalent software for interface prototyping, Google Drive for documentation, GitHub for version control, and Discord for coordination. Quality assurance will include peer review, requirements traceability and conformance checking, accessibility review, and verification of the simulated beacon interaction. Full unit, integration, system, and automated testing are outside Phase I.

4.2. Software Documentation

Phase I documentation will focus on course-required requirements work products rather than production-oriented artifacts such as deployment guides or formal test plans. Documentation and prototype materials must remain consistent with the approved world assumptions, including indoor-only navigation, room-to-room movement, simulated beacon signals, and emergency dial-out.

Work Product	Point Person	Standard / Template	Work Period	Review	Target	Distribution
Preliminary Project Plan	Alex Shaw	Course Project Plan template	9/10–9/20	All	9/20	Drive, GitHub, course submission
Interface Mock-up / Prototype	David Del Val Fraire	Course requirements; accessibility conventions	9/21–10/2	Johanna, Seth	10/4	Drive, GitHub

User Manual	Johanna Siebol	Course requirements; approved mock-up	9/28–10/3	David	10/4	Drive, GitHub
WRS Document	Danbi Kim	Course WRS requirements	9/21–10/4	David, Seth	10/6	Drive, GitHub
AS-IS / TO-BE Scenarios	Johanna Siebol	Required scenario format	9/28–10/7	Danbi, Alex	10/8	Drive, presentation
Phase I Demo Slides	Johanna Siebol	Course presentation requirements	10/5–10/7	Alex, Danbi	10/8	Drive, presentation
Phase I Peer Review	Alex Shaw	Course peer-review format	10/9–10/12	All	10/12	Course submission
GenAI Usage / Verification Report	Seth Trimberger	Course GenAI requirements	Ongoing –10/17	Alex	10/18	Drive, final submission
Phase I Final Submission	Seth Trimberger	Course submission requirements	10/12–10/17	All	10/18	Course submission, GitHub

Drafts will be maintained in Google Drive, with baselined artifacts retained in GitHub. The mock-up, WRS, scenarios, and user manual should all describe the same beacon-based indoor navigation model to maintain cross-document consistency.